



AW300H Butt Weld Angle Valve

INTRODUCTION

These low pressure drop refrigerant shut-off valves are designed to be butt-welded directly to steel piping, thereby eliminating potential leaky flanges or threaded joints and simplifying installation. Forged and cast steel bodies are lightweight, yet have substantial wall thickness to help overcome corrosion loss. The cast steel body is rigid, reducing the potential for seat leakage due to flexing of the valve body under abnormal conditions. The “heart” of Hansen shut-off valves is the patented non-leak seal plus packing design. This seal design is used exclusively on Hansen shut-off valves and virtually eliminates stem seal leakage.

APPLICATIONS

Typical uses include:

Ammonia refrigeration system suction, liquid, discharge, recirculating liquid, hot gas, thermosyphon, and oil lines, using handwheel or seal cap models.

Steel pipe portions of halocarbon, commercial, industrial, and air conditioning systems, using seal cap models.

Valves are also available for other fluids such as propane, propylene, and glycol with compatible seals. Contact factory with specific fluid and application details for written approval.

The low friction, no leak stem seal design permits the use of chain actuation for crossover applications without the need to retighten the packing.

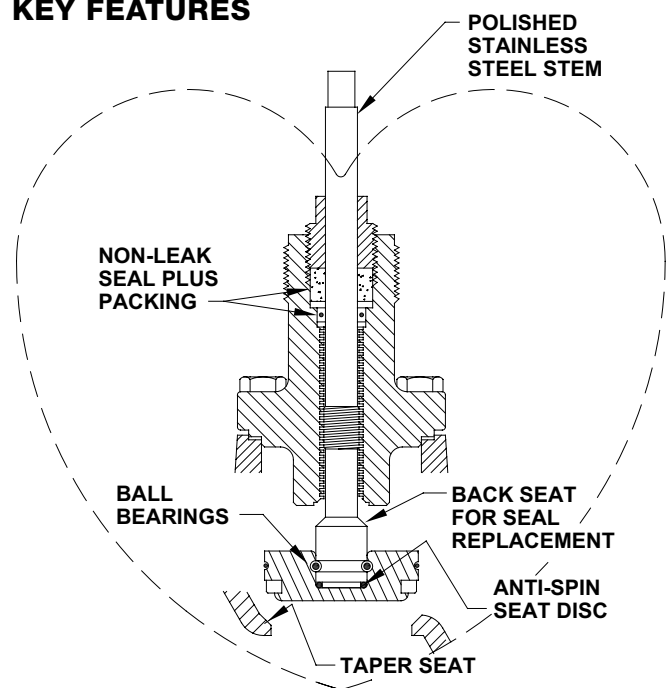
Specifications, Applications, Service Instructions & Parts

BUTT WELD SHUT-OFF VALVES

**1/2" through 18"
(13mm through 450mm)
Globe & Angle
for Refrigerants**

CRN CE
Canadian Registration

KEY FEATURES



ADDITIONAL FEATURES

Globe and angle valves available.

Low pressure drop design

Interchangeable handwheel or seal cap.

Teflon seat disc.

Also available as Hand Expansion Valve.

Chain actuators available (1½" or 40mm and larger).

Suitable for R717 (ammonia), R22, R134a, R507a, R404a, R744 (CO₂) and other Hansen approved refrigerants.

600 psig (41 bar g) SWP valves are available.

Extended neck available up through 4" (100mm) for additional insulation clearance.

MATERIAL SPECIFICATIONS

Body: ½" to 1¼" (13mm–32mm), forged steel, ASTM A105; 1½" to 18" (40mm–450mm), cast steel, ASTM A-352, Grade LCB

Bonnet: Ductile iron, ½" to 8" (13mm–200mm) ASTM A-536; 10" to 18" (250mm–450mm) ASME SA-395

Stem: Polished stainless steel

Disc Holder: Plated steel

Seat Disc: Retained PTFE Teflon

Ball Bearings: Stainless steel

Packing Nut: Zinc plated steel

Stem Packing: Graphite composite plus neoprene O-ring in series

Handwheel: ½" to 1¼" (13mm–32mm) Zinc plated alloy; 1½" to 6" (40mm–150mm) Zinc plated iron alloy; 8" to 18" (200mm–450mm) cast aluminum

Seal Cap: ½" to 1¼" (13mm–32mm) Glass-filled polymer, safety vented; 1½" to 4" (40mm–100mm) Zinc plated steel; 5" to 18" (125mm–450mm) cast aluminum

Operating Temperature:

Standard, –60°F to 240°F (–51°C to 115°C), temperatures below –60°F (–51°C) to –76°F (–60°C) at coincident pressures

Optional, –20°F to 300°F (–29°C to 149°C) contact factory

Safe Working Pressure: 400 psig (28 bar g) standard, 600 psig (41 bar g) upon request

Connection Dimensions: ½" to 1 ½" Match Schedule 80 pipe, 2" to 10" Match Schedule 40 pipe, 12" & Up Match Standard Weight Pipe

ADVANTAGES

Compared to conventional stem seal designs, Hansen's patented no leak seal-plus-stem-packing virtually eliminates stem seal leakage. In addition, the stem is polished to avoid packing wear. The precision stem threads avoid stem wobble and the packing nut is close fitting to remove dirt particles and frost from the stem. This results in infrequent maintenance or tightening and almost no refrigerant loss.

Hansen valves are designed with anti-spin seat discs. High velocity gas can sometimes cause ordinary seat discs to fail over time. Hansen adds a snubber to prevent the seat disc from freely rotating when valve is in the open position.

Compared to threaded valves, Hansen welding valves eliminate the chance of future leaks at pipe threads. In addition, a butt weld pipe-to-valve body joint eliminates the inherent weakness and corrosion vulnerability of the threaded portion of pipe immediately adjacent to a screwed valve body or flange.

Compared to ball valves, Hansen welding valves have no threaded or flanged-gasket pipe joints. Hansen valves also will not open or close so rapidly as to cause severe pipe shock due to sudden change in liquid velocity. In addition, ball valve stem packing typically cannot be replaced while the valve is in a pressurized line. The stem packing on these Hansen valves can be replaced while the valve is in a pressurized line. Also, dirt or damage to ball valve Teflon seats cannot be overcome by greater stem closing forces, an advantage of Hansen valves.

INSTALLATION

All Hansen welded valves can be installed in horizontal or vertical pipe lines. Stems are normally installed horizontally, but, depending on the application, stems may be installed vertically. Inlet pressure or direction of flow for all valve sizes should normally be under valve seat disc. However, to avoid installing an angle valve with the stem down, it is better to install the valve with the normal flow opposite the direction of the arrow. For pressure differences exceeding approximately 125 psi (9 bar), valves 8" (200mm) and larger should have inlet pressure above valve disc, using a small bypass valve for preopening equalization.

Care must be taken when handling and installing large valves. Proper lifting devices and safety precautions must be observed.

A valve should have its bonnet assembly removed before welding. This reduces weight during welding, protects the Teflon seat disc from welding sparks, and facilitates cleaning of welding debris from the body interior prior to valve operation. The valve stem should be several turns open when removing and replacing the bonnet assembly. The Teflon seat disc should be protected when outside of the valve. Do not allow the Teflon seat to bump the valve body when removing it or stand the bonnet on the seat disc.

Where it is necessary or standard practice to weld a valve into the line without bonnet removal, the stem should be opened several turns to prevent heat damage to the seat disc. Extra care should be taken when welding angle valves without disassembly to avoid welding sparks striking the seat from the outlet weld connection.

Use of welding rings is recommended. They help alignment, control the gap for full penetration welding, and reduce the potential of welding debris entering the system. Welds should be annealed as necessary in accordance with good practice. Painting of valves and welds is recommended for corrosion protection. Pipe covering, where applied, should have a proper moisture barrier.

Before putting valves into service, all pipe weld connections, valve seats, bonnet seals, and stem seals should be tested for leaks at pressure levels called for in appropriate codes.

Shut-off valves leading to the atmosphere must not be left unsupervised and must be plugged or capped to prevent corrosion inside the valve as well as leakage due to seat expansion, vibration, pressure shock, or improper opening. Where hand wheel valves are subject to excessive vibration or liquid shock loads, the packing nut should be tightened to prevent handwheel vibration from potentially rotating valve stem. The valve seat should be cracked open to prevent hydrostatic expansion between the valve and the cap. A bleed valve and gauge should be installed to provide a means of monitoring and pump-out. Valves should never directly feed a water tank because of potential internal corrosion or seat opening caused by vibration.

INSULATION

Conventional valve-shaped block insulation can occasionally be used for both angle and globe valves. However, fabricated insulation shapes are recommended. If not available locally, Hansen can recommend a source of high quality, economical valve insulation. For exterior valve insulation dimensions refer to Sales Drawing 5002-28.

EXTENDED NECK VALVES

Hansen bonnets with extended necks, 1/2" thru 4" (13mm–100mm), meet or exceed the recommended low temperature insulation thickness dimensions of the IAR Piping Handbook. In addition, the stem and packing nut trim are upgraded to 316 stainless steel materials. Stainless steel trim is recommended for environments where severe corrosive conditions exist. (Add an "E" suffix to the model number, for example AWE200C or GWE150H.)

SIZING GUIDE

These flow capacity recommendations are not affected by the length of the pipe line. These are approximate optimum sizes based on power costs versus the investment costs of piping and its total installed cost. Piping sized to this capacity will have a 1°F (0.6°C) pressure drop for the following equivalent lengths:

suction lines 700 diameters

discharge lines 1400 diameters

liquid lines 2400 diameters

Example: 275 feet (84m) of 3" (80mm) pipe and equivalent fittings amount to 1100 diameters, pressure drop for a suction line handling 81.5 tons (287 kW) at 20°F (–7°C) is 1100/700 times 1°F (0.6°C) drop, equals 1.6°F (1°C) or 1.8 psi (0.12 bar).

Example: Hansen valves have about 145 diameters of equivalent flow resistance, or 145/700 = 0.2°F (0.12°C) of equivalent pressure drop at the suction line capacities shown for a valve in a suction line.

The rationale for the vapor line sizing was developed by William V. Richards in two papers: "Refrigerant Vapor Line Sizing Not Dependent of Length," 16th International Congress of Refrigeration, IIR, Paris, 1983; and "Practical Pipe Sizing for Refrigerant Vapor Lines," Sixth Annual Meetings, IAR, San Francisco, 1984.

FLOW COEFFICIENTS

NOMINAL SIZE		ANGLE				GLOBE			
INCH	MM	FLOW COEFFICIENT		EQUIVALENT LENGTH		FLOW COEFFICIENT		EQUIVALENT LENGTH	
		Cv	Kv	FEET	METERS	Cv	Kv	FEET	METERS
1/2"	13	6.0	5.2	3.7	1.1	4.0	3.5	8.4	2.6
3/4"	20	9.0	7.8	8.4	2.6	8.0	6.9	11	3.2
1"	25	19	16	7.2	2.2	15	13	12	3.5
1¼"	32	21	18	27	8.2	16	14	46	14
1½"	40	53	46	10	3.0	41	35	16	5.0
2"	50	80	69	23	7.0	67	58	33	10
2½"	65	173	150	13	3.8	163	141	14	4.3
3"	80	205	177	28	8.5	195	169	31	9.3
4"	100	320	277	47	14.3	290	251	57	17
5"	125	600	519	44	13.3	575	497	47	14
6"	150	820	709	61	18.5	790	683	66	20
8"	200	1440	1240	82	25.1	1380	1190	90	27
10"	250	2450	2120	93	28.4	2350	2030	101	31
12"	300	3400	2940	119	36.2	3270	2830	128	39
14"	350	4600	3980	113	34.4	4350	3770	126	38
16"	400	5640	4880	155	47.3	–	–	–	–
18"	450	6500	5620	222	67.6	–	–	–	–

**FLOW CAPACITIES
PIPING AND VALVE SIZING GUIDE FOR AMMONIA**

SERVICE	CONDITIONS				CAPACITIES (MM)									
	TEMPERATURE		PRESSURE		1/2" (13)		3/4" (20)		1" (25)		1¼" (32)		1½" (40)	
	°F	(°C)	PSIG	(BAR)	TONS	(kW)	TONS	(kW)	TONS	(kW)	TONS	(kW)	TONS	(kW)
Suction Lines Single Stage Compressor	+20	(-6.7)	33.5	(2.3)	–	–	–	–	8.6	(30)	15.8	(56)	21.3	(75.0)
	0	(-17.8)	15.7	(1.1)	–	–	–	–	5.7	(20)	10.4	(37)	13.9	(49.0)
Suction Lines Booster	-20	(-28.9)	3.6	(0.25)	–	–	–	–	4.2	(15)	7.4	(26)	10.3	(36.0)
	-40	(-40)	8.7"	(0.7)	–	–	–	–	–	–	4.4	(15)	6.3	(22.0)
Liquid Overfeed Return Lines (4X)	+20	(-6.7)	33.5	(2.3)	–	–	–	–	5.0	(18)	9.1	(32)	12.3	(43.0)
	0	(-17.8)	15.7	(1.1)	–	–	–	–	3.4	(12)	6.3	(22)	8.5	(30.0)
	-20	(-28.9)	3.6	(0.25)	–	–	–	–	2.2	(8.0)	4.0	(14)	5.5	(19.0)
	-40	(-40)	8.7"	(0.7)	–	–	–	–	–	–	2.4	(8.0)	3.4	(12.0)
Hot Gas Feed	+70	(+21.1)	114	(7.9)	2.2	(8)	4.3	(15)	7.3	(26)	14.1	(50)	19.6	(69.0)
Hot Gas Main	+70	(+21.1)	114	(7.9)	4.4	(15)	8.6	(30)	14.7	(52)	28.1	(99)	39.2	(138)
Compressor Discharge	+86	(+30)	155	(10.7)	–	–	–	–	12.6	(44)	24.1	(85)	33.6	(118)
Condenser Drains	+86	(+30)	–	–	6.0	(21)	14.5	(51)	24.0	(84)	50.0	(176)	77.0	(271)
Liquid Mains	+86	(+30)	–	–	28.3	(100)	53.1	(187)	90.8	(320)	143	(503)	202	(711)
Liquid Feed Branch	+86	(+30)	–	–	54.9	(193)	103	(363)	176	(620)	277	(975)	392	(1380)
Liquid Overfeed Supply (4X)	+10	(-12.2)	–	–	9.0	(32)	17.0	(60)	29.0	(102)	46.0	(162)	65.0	(229)

SERVICE	CONDITIONS		CAPACITIES (MM)				
	TEMPERATURE °F (°C)	PRESSURE PSIG (BAR)	2" (50) TONS (kW)	2 1/2" (65) TONS (kW)	3" (80) TONS (kW)	4" (100) TONS (kW)	5" (125) TONS (kW)
Suction Lines Single Stage Compressor	+20 (-6.7)	33.5 (2.3)	35.7 (126)	51.1 (180)	81.5 (287)	146 (513)	235 (826)
	0 (-17.8)	15.7 (1.1)	22.7 (80.0)	34.0 (120)	53.9 (190)	94.6 (333)	156 (549)
Suction Lines Booster	-20 (-28.9)	3.6 (0.25)	16.8 (59.0)	24.8 (87.0)	39.7 (140)	68.9 (242)	114 (401)
	-40 (-40)	8.7" (0.7)	9.9 (35.0)	14.4 (51.0)	23.4 (82.3)	40.8 (143)	66.8 (235)
Liquid Overfeed Return Lines (4X)	+20 (-6.7)	33.5 (2.3)	20.6 (72.4)	29.4 (103)	47.0 (165)	84.3 (296)	135 (475)
	0 (-17.8)	15.7 (1.1)	13.6 (48.0)	20.5 (72.0)	32.4 (114)	56.8 (200)	93.8 (330)
	-20 (-28.9)	3.6 (0.25)	8.9 (31.3)	13.1 (46.0)	21.0 (74.0)	36.4 (128)	60.5 (213)
	-40 (-40)	8.7" (0.7)	5.4 (19.0)	7.90 (28.0)	12.8 (45.0)	22.4 (79.0)	36.7 (129)
Hot Gas Feed Hot Gas Main	+70 (+21.1)	114 (7.9)	36.5 (128)	53.0 (186)	82.5 (290)	145 (510)	231 (812)
	+70 (+21.1)	114 (7.9)	73.0 (257)	106 (373)	165 (580)	290 (1020)	463 (1628)
Compressor Discharge	+86 (+30)	155 (10.7)	62.6 (220)	90.3 (318)	142 (499)	249 (876)	397 (1396)
Condenser Drains	+86 (+30)	- -	140 (492)	220 (774)	375 (1319)	740 (2602)	1320 (4642)
Liquid Mains	+86 (+30)	- -	454 (1597)	657 (2311)	1031 (3626)	1808 (6358)	2886 (10150)
Liquid Feed Branch	+86 (+30)	- -	881 (3098)	1273 (4477)	1999 (7030)	3506 (12330)	5596 (19680)
Liquid Overfeed Supply (4X)	+10 (-12.2)	- -	144 (506)	208 (732)	327 (1150)	573 (2015)	915 (3218)

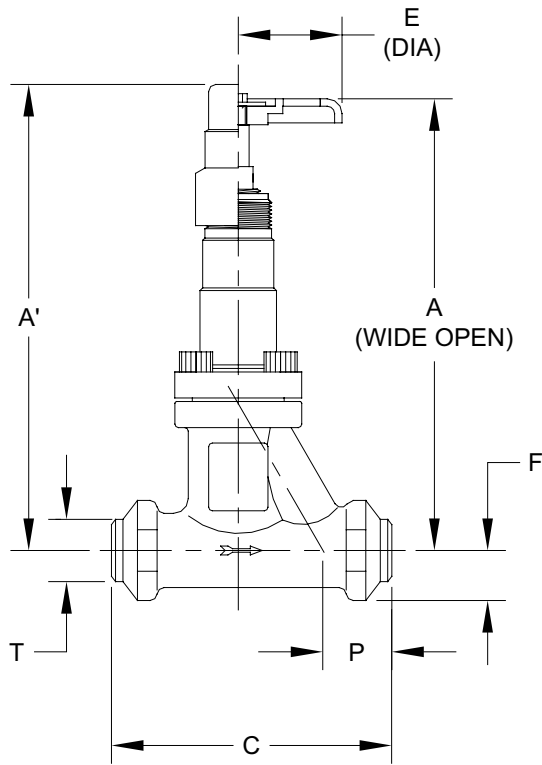
**FLOW CAPACITIES
PIPING AND VALVE SIZING GUIDE FOR AMMONIA**

SERVICE	CONDITIONS		CAPACITIES (MM)							
	TEMPERATURE °F (°C)	PRESSURE PSIG (BAR)	6" (150) TONS (kW)	8" (200) TONS (kW)	10" (250) TONS (kW)	12" (300) TONS (kW)	14" (350) TONS (kW)			
Suction Lines Single Stage Compressor	+20 (-6.7)	33.5 (2.3)	343 (1206)	628 (2209)	1020 (3387)	1490 (5240)	1821 (6404)			
	0 (-17.8)	15.7 (1.1)	225 (791.0)	414 (1456)	662 (2328)	946 (3327)	1156 (4065)			
Suction Lines Booster	-20 (-28.9)	3.6 (0.25)	165 (580.0)	306 (1076)	486 (1709)	709 (2494)	867 (3049)			
	-40 (-40)	8.7" (0.7)	97.8 (344.0)	181 (637.0)	288 (1013)	419 (1474)	512 (1801)			
Liquid Overfeed Return Lines (4X)	+20 (-6.7)	33.5 (2.3)	198 (696.0)	362 (1273)	587 (2064)	856 (3010)	1046 (3679)			
	0 (-17.8)	15.7 (1.1)	135 (475.0)	249 (876.0)	398 (1400)	568 (1998)	694 (2441)			
	-20 (-28.9)	3.6 (0.25)	87.5 (308.0)	162 (570.0)	256 (900)	375 (1319)	458 (1611)			
	-40 (-40)	8.7" (0.7)	53.7 (189.0)	100 (352.0)	158 (556)	232 (816)	284 (999)			
Hot Gas Feed Hot Gas Main	+70 (+21.1)	114 (7.9)	338 (1189)	595 (2093)	949 (3338)	1377 (4843)	1683 (5919)			
	+70 (+21.1)	114 (7.9)	673 (2367)	1190 (4185)	1898 (6675)	2754 (9685)	3366 (11838)			
Compressor Discharge	+86 (+30)	155 (10.7)	580 (2040)	1021 (3591)	1629 (5729)	2363 (8310)	2888 (10157)			
Condenser Drains	+86 (+30)	- -	2030 (7139)	4200 (14771)	- -	- -	- -			
Liquid Mains	+86 (+30)	- -	4218 (14834)	- -	- -	- -	- -			
Liquid Feed Branch	+86 (+30)	- -	8179 (28764)	- -	- -	- -	- -			
Liquid Overfeed Supply (4X)	+10 (-12.2)	- -	1337 (4702)	- -	- -	- -	- -			

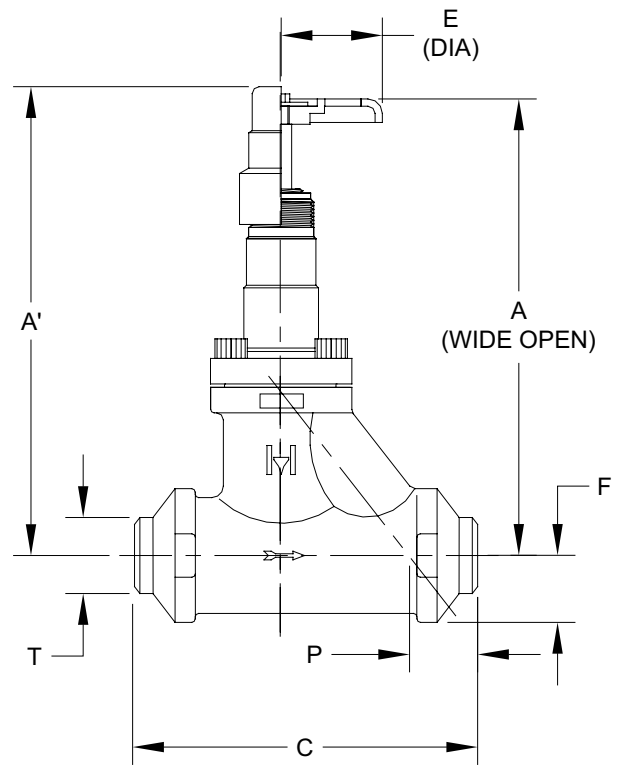
**FLOW CAPACITIES
PIPING AND VALVE SIZING GUIDE FOR AMMONIA**

SERVICE	CONDITIONS		CAPACITIES (MM)			
	TEMPERATURE °F (°C)	PRESSURE PSIG (BAR)	16" (400) TONS (kW)	18" (450) TONS (kW)		
Suction Lines Single Stage Compressor	+20 (-6.7)	33.5 (2.3)	2310 (8124)	2663 (9365)		
	0 (-17.8)	15.7 (1.1)	1684 (5922)	1941 (6826)		
Suction Lines Booster	-20 (-28.9)	3.6 (0.25)	1162 (4087)	1339 (4709)		
	-40 (-40)	8.7" (0.7)	692 (2434)	798 (2806)		
Liquid Overfeed Return Lines (4X)	+20 (-6.7)	33.5 (2.3)	1258 (4424)	1450 (5099)		
	0 (-17.8)	15.7 (1.1)	944 (3320)	1088 (3826)		
	-20 (-29)	3.6 (0.25)	614 (2159)	708 (2490)		
	-40 (-40)	8.7" (0.7)	381 (1340)	439 (1544)		
Hot Gas Feed Hot Gas Main	+70 (+21)	114 (7.9)	- -	- -		
	+70 (+21)	114 (7.9)	- -	- -		
Compressor Discharge	+86 (+30)	155 (10.7)	3883 -	- -		

GLOBE INSTALLATION DIMENSIONS
½" THROUGH 1¼" (13MM THROUGH 32MM)



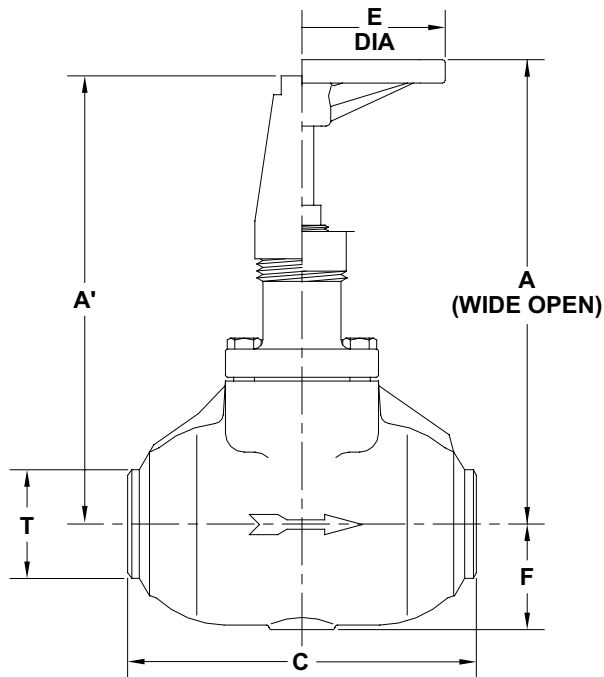
1/2" AND 3/4"
(13MM AND 20MM)



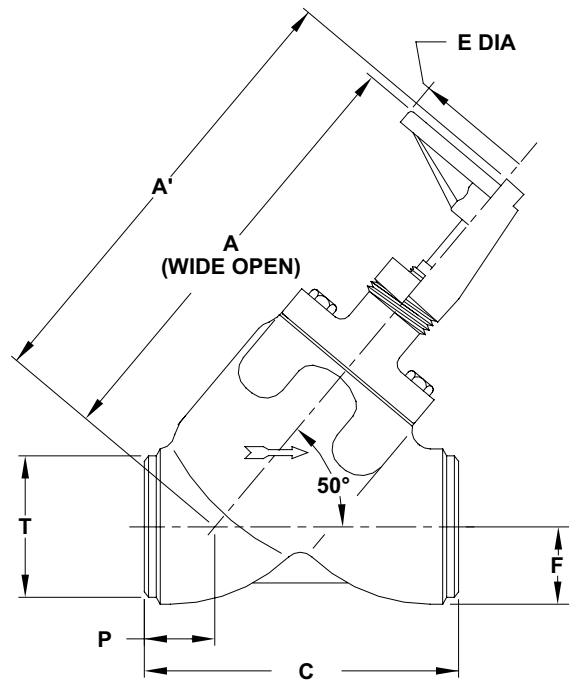
1" AND 1¼"
(25MM AND 32MM)

SIZE (MM)	A	A'	C	E	F	P	T	ADD TO A, A' EXT NECK
1/2" (13)	7.62" (194)	7.86" (200)	4.73" (120)	3.5" (89)	0.84" (21)	1.17" (30)	0.84" (21)	1.8" (45)
3/4" (20)	7.62" (194)	7.86" (200)	4.73" (120)	3.5" (89)	0.84" (21)	1.17" (30)	1.05" (27)	1.8" (45)
1" (25)	7.88" (200)	8.09" (206)	5.93" (150)	3.50" (89)	1.15" (29)	1.17" (30)	1.31" (33)	1.8" (45)
1¼" (32)	7.88" (200)	8.09" (206)	5.93" (150)	3.50" (89)	1.15" (29)	1.17" (30)	1.66" (42)	1.8" (45)

GLOBE INSTALLATION DIMENSIONS
1½" THROUGH 14" (40MM THROUGH 350MM)



1½" – 3", 4" EXT NECK
(40MM – 80MM), (100MM)

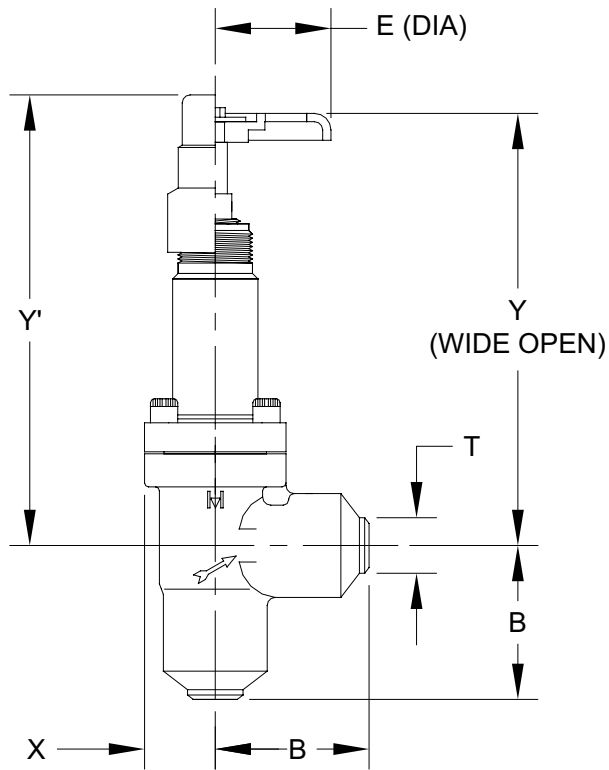


4" – 14"
(100MM – 350MM)

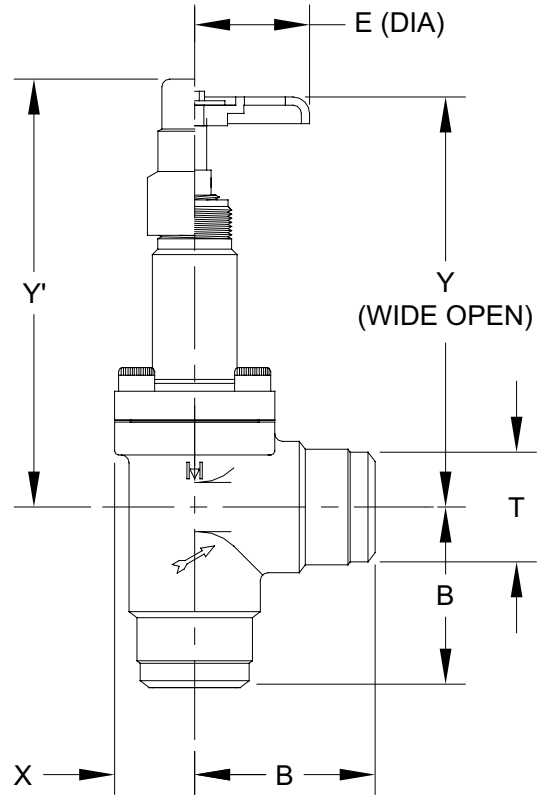
SIZE (MM)	A	A'	C	E	F	P	T	ADD TO A, A' EXT NECK
1½" (40)	8.88" (225)	9.13" (232)	7.25" (184)	4.25" (108)	2.12" (54)	–	1.90" (48)	2.85" (72)
2" (50)	8.88" (225)	9.13" (232)	7.25" (184)	4.25" (108)	2.13" (54)	–	2.38" (60)	2.85" (72)
2½" (65)	12.13" (308)	11.75" (298)	9.25" (235)	7.63" (194)	2.75" (70)	–	2.88" (73)	3.64" (92)
3" (80)	12.13" (308)	11.75" (298)	9.25" (235)	7.63" (194)	2.75" (70)	–	3.50" (89)	3.64" (92)
4" (100)	13.75" (349)	14.25" (362)	10.00" (254)	7.63" (194)	2.50" (64)	2.50" (64)	4.50" (114)	–
4" EXT NECK* (100)	13.12" (333)	12.12" (308)	12.50" (318)	7.63" (194)	3.75" (95)	–	4.50" (114)	–
5" (125)	21.38" (543)	21.63" (549)	11.65" (396)	10.00" (254)	3.00" (76)	2.65" (67)	5.56" (141)	–
6" (150)	22.10" (561)	22.15" (563)	13.26" (337)	10.00" (254)	3.50" (89)	2.73" (69)	6.63" (168)	–
8" (200)	29.10" (739)	29.51" (750)	18.10" (459)	16.00" (406)	4.61" (117)	3.98" (101)	8.63" (219)	–
10" (250)	35.29" (896)	36.16" (918)	23.38" (594)	16.00" (406)	5.81" (148)	5.19" (132)	10.75" (273)	–
12" (300)	42.16" (1071)	46.39" (1178)	29.20" (742)	20.00" (508)	7.00" (178)	6.27" (159)	12.75" (324)	–
14" (350)	47.55" (1208)	49.66" (1261)	32.86" (835)	20.00" (508)	7.63" (194)	7.23" (184)	14.00" (356)	–

* 4" (100mm) Extended Neck is "T" style body only.

ANGLE INSTALLATION DIMENSIONS
½" THROUGH 1¼" (13MM THROUGH 32MM)



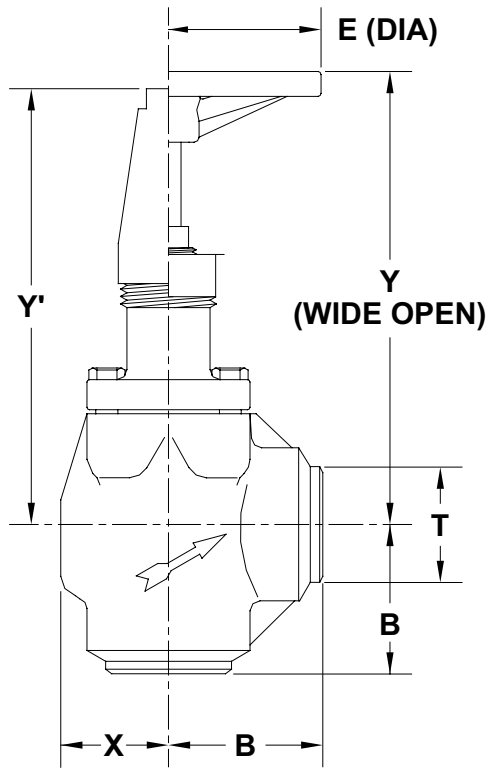
1/2" AND 3/4"
(13MM AND 20MM)



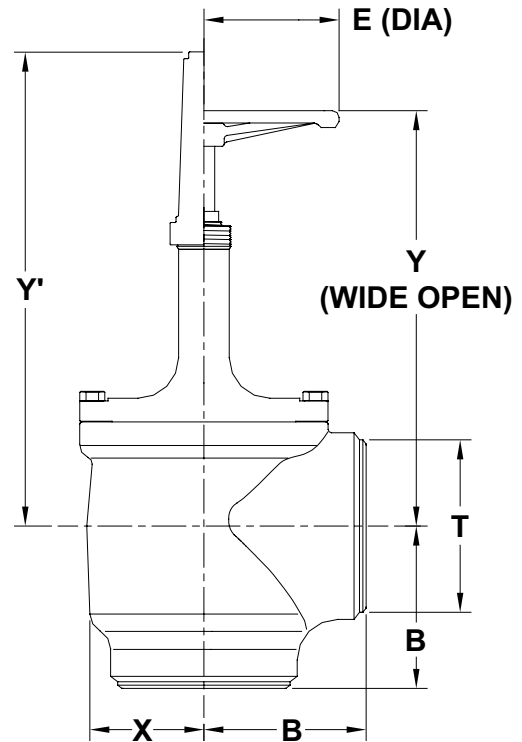
1" AND 1¼"
(25MM AND 32MM)

SIZE (MM)	B	E	T	X	Y	Y'	ADD TO Y, Y' EXT NECK
1/2" (13)	2.33" (59)	3.50" (89)	0.84" (21)	1.07" (27)	6.46" (164)	6.70" (170)	1.80" (45)
3/4" (20)	2.33" (59)	3.50" (89)	1.05" (27)	1.07" (27)	6.46" (164)	6.70" (170)	1.80" (45)
1" (25)	2.75" (70)	3.50" (89)	1.31" (33)	1.22" (31)	6.24" (158)	6.52" (166)	1.80" (45)
1¼" (32)	2.75" (70)	3.50" (89)	1.66" (42)	1.22" (31)	6.24" (158)	6.52" (166)	1.80" (45)

ANGLE INSTALLATION DIMENSIONS
1½" THROUGH 18" (40MM THROUGH 450MM)



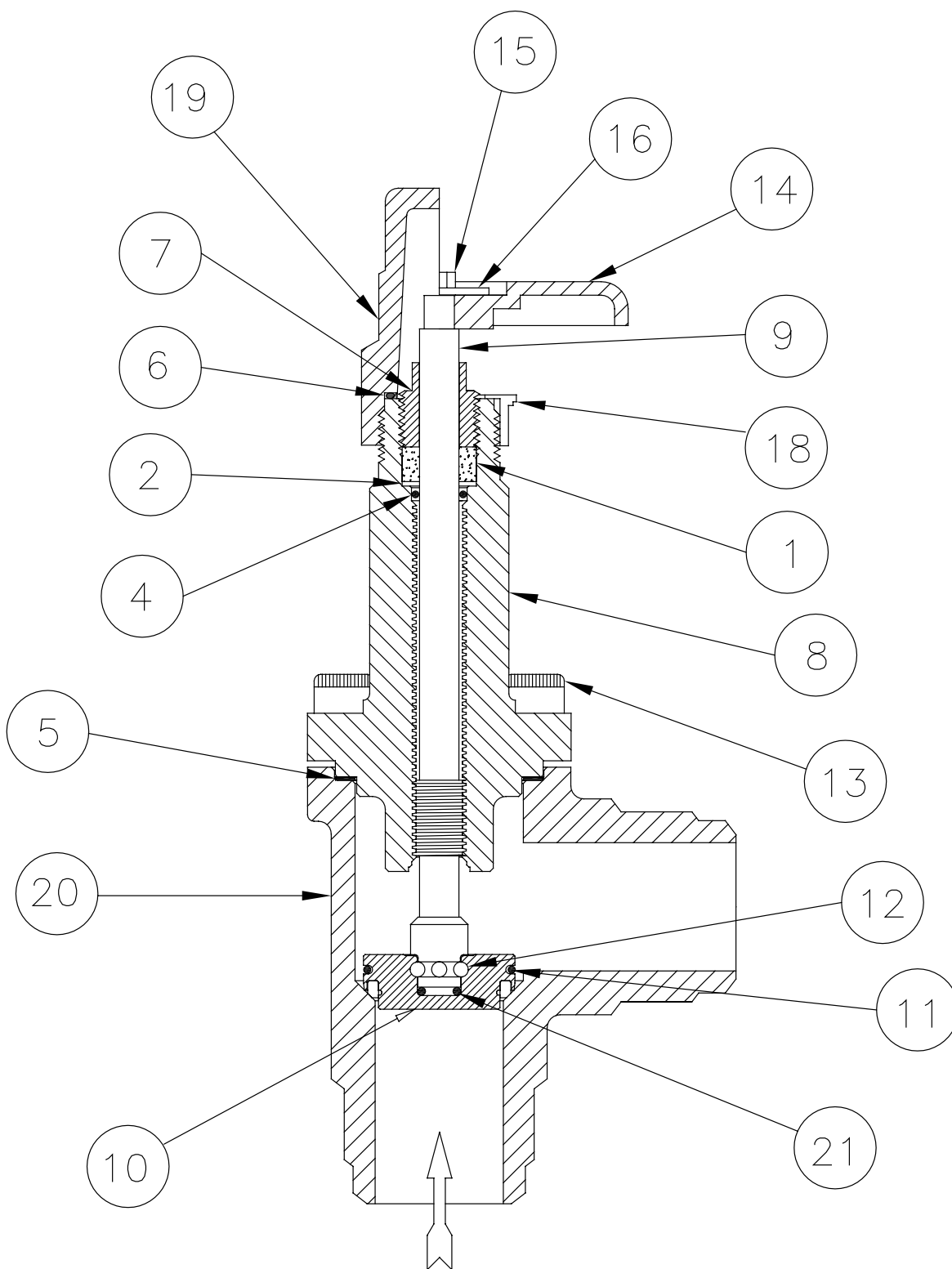
1½" – 8"
(40MM – 200MM)



10" – 18"
(250MM – 450MM)

SIZE (MM)	B	E	T	X	Y	Y'	ADD TO Y, Y' EXT NECK
1½" (40)	3.38" (86)	4.25" (108)	1.90" (48)	1.88" (48)	7.75" (197)	8.00" (203)	2.85" (72)
2" (50)	3.38" (86)	4.25" (108)	2.38" (60)	1.88" (48)	7.75" (197)	8.00" (203)	2.85" (72)
2½" (65)	3.38" (86)	7.63" (194)	2.88" (73)	2.38" (60)	11.00" (280)	10.63" (270)	3.64" (92)
3" (80)	3.38" (86)	7.63" (194)	3.50" (89)	2.38" (60)	11.00" (280)	10.63" (270)	3.64" (92)
4" (100)	3.88" (98)	7.63" (194)	4.50" (114)	3.00" (76)	11.00" (280)	10.63" (270)	3.65" (93)
5" (125)	4.47" (131)	10.00" (254)	5.56" (141)	3.47" (88)	16.42" (417)	16.62" (422)	–
6" (150)	5.15" (131)	10.00" (254)	6.63" (168)	3.85" (98)	16.55" (420)	16.78" (426)	–
8" (200)	6.90" (175)	16.00" (406)	8.63" (219)	6.11" (155)	22.12" (562)	22.52" (572)	–
10" (250)	9.44" (240)	16.00" (406)	10.75" (273)	6.63" (168)	26.47" (672)	27.34" (694)	–
12" (300)	12.00" (305)	20.00" (508)	12.75" (324)	8.66" (220)	30.34" (783)	33.86" (860)	–
14" (350)	12.97" (329)	20.00" (508)	14.00" (356)	9.20" (234)	34.92" (887)	36.91" (938)	–
16" (400)	13.16" (334)	30.00" (752)	16.00" (406)	10.25" (260)	59.45" (1510)	45.60" (1158)	–
18" (450)	14.42" (366)	30.00" (752)	18.00" (457)	11.55" (293)	62.40" (1585)	44.50" (1130)	–

1/2" THROUGH 1 1/4" (13 MM THROUGH 32 MM) BUTT WELD SHUT-OFF VALVES



NOTE: Angle valve shown; kits are the same for globe valves.

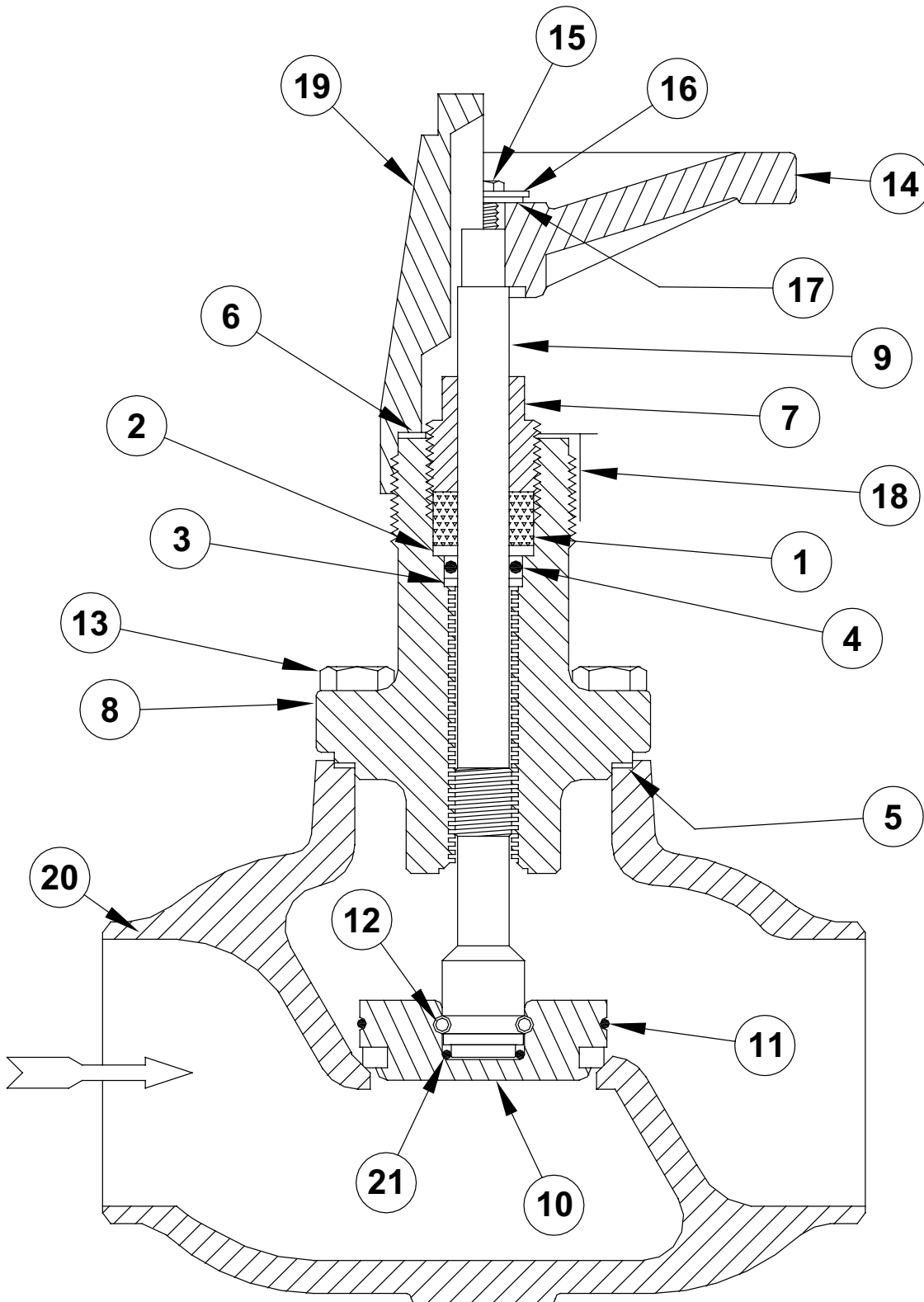
REPLACEMENT PARTS – BUTT WELD SHUT-OFF VALVES

1/2" THROUGH 1 1/4" (13MM THROUGH 32MM)

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 1/2", 3/4", 1", 1-1/4" (13mm, 20mm, 25mm, 32mm)		50-1040
	Gasket Kit, Extended Neck 1/2", 3/4", 1", 1-1/4" (13mm, 20mm, 25mm, 32mm)		50-1171
	<i>Above kits consist of:</i>		
1	Stem Packing	1	
2	Stem Washer	1	
4	Stem O-Ring	1	
5	Bonnet Gasket	1	
6	Seal Cap O-Ring	1	
7	Packing Nut	1	
7b	Packing Nut, Extended Neck	1	
21	O-Ring, Anti-Spin	1	
	Bonnet Assembly Kit 1/2", 3/4" (13mm, 20mm)		50-1094
	Bonnet Assembly Ext Neck Kit 1/2", 3/4" (13mm, 20mm)		50-1172
	Bonnet Assembly Kit 1", 1-1/4" (25mm, 32mm)		50-1095
	Bonnet Assembly Ext Neck Kit 1", 1-1/4" (25mm, 32mm)		50-1174
	<i>Above kits consist of:</i>		
8	Bonnet	1	
9	Stem	1	
9b	Stem, Extended Neck	1	
10	Disc Assembly	1	
11	Ball Retainer	1	
12	Balls	10	
13	Bonnet Cap Screw	4	
	Gasket Kit	1	
	Handwheel Kit		50-1005
	<i>Above kits consist of:</i>		
14	Handwheel	1	
15	Screw	1	
16	Nameplate	1	
18	Bonnet Thread Cap	1	
	Seal Cap Kit		50-1036
	<i>Above kits consist of:</i>		
6	Seal Cap O-Ring	1	
19	Seal Cap	1	
20	Body	1	

NOTE: A plate steel cap is available to order, specify part number 75-1014.

1½" THROUGH 4" (40MM THROUGH 100MM) – BUTT WELD SHUT-OFF VALVES



NOTE: Globe valve shown; kits are the same for angle valves.

REPLACEMENT PARTS – BUTT WELD SHUT-OFF VALVES
1½" THROUGH 4" (40MM THROUGH 100MM)

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 1-1/2", 2" (40mm, 50mm) Gasket Kit Ext Neck 1-1/2", 2" (40mm, 50mm) Gasket Kit 2-1/2", 3" (65mm, 80mm) Gasket Kit Ext Neck 2-1/2", 3" (65mm, 80mm) Gasket Kit 4" (100mm) Gasket Kit Ext Neck 4" (100mm) <i>Above kits consist of:</i>		50-1023 50-1164 50-1043 50-1166 50-1065 50-1203
1	Stem Packing	1	
2	Stem Washer	1	
3	Back-up Washer	1	
4	Stem O-Ring	1	
5	Bonnet Gasket	1	
6	Seal Cap Gasket	1	
7	Packing Nut	1	
21	O-Ring, Anti-Spin	1	
	Bonnet Assembly Kit 1-1/2", 2" (40mm, 50mm) Bonnet Assembly Ext Neck Kit 1-1/2", 2" (40mm, 50mm) Bonnet Assembly Kit 2-1/2", 3" (65mm, 80mm) Bonnet Assembly Ext Neck Kit 2-1/2", 3" (65mm, 80mm) Bonnet Assembly Kit 4" Bonnet Assembly Ext Neck Kit 4" (100mm) <i>Above kits consist of:</i>		50-1024 50-1165 50-1044 50-1167 50-1066 50-1184
8	Bonnet	1	
8b	Bonnet Ext Neck	1	
9	Stem	1	
9b	Stem Ext Neck	1	
13	Bonnet Cap Screw	4	
	Gasket Kit	1	
10	Disc Assembly Kit	1	
11	Ball Retainer	1	
12	Balls	1	
	Handwheel Kit 1-1/2", 2" (40mm, 50mm) Handwheel Kit 2-1/2", 3", 4" (65mm, 80mm, 100mm) <i>Above kits consist of:</i>		50-1026 50-1037
14	Handwheel	1	
15	Screw	1	
16	Nameplate	1	
17	Support Washer	1	
18	Bonnet Thread Cap	1	
	Seal Cap Kit 1-1/2", 2" (40mm, 50mm) Seal Cap Kit 2-1/2", 3", 4" (65mm, 80mm, 100mm) <i>Above kits consist of:</i>		50-1027 50-1038
6	Seal Cap Gasket	1	
19	Seal Cap	1	
20	Body	1	

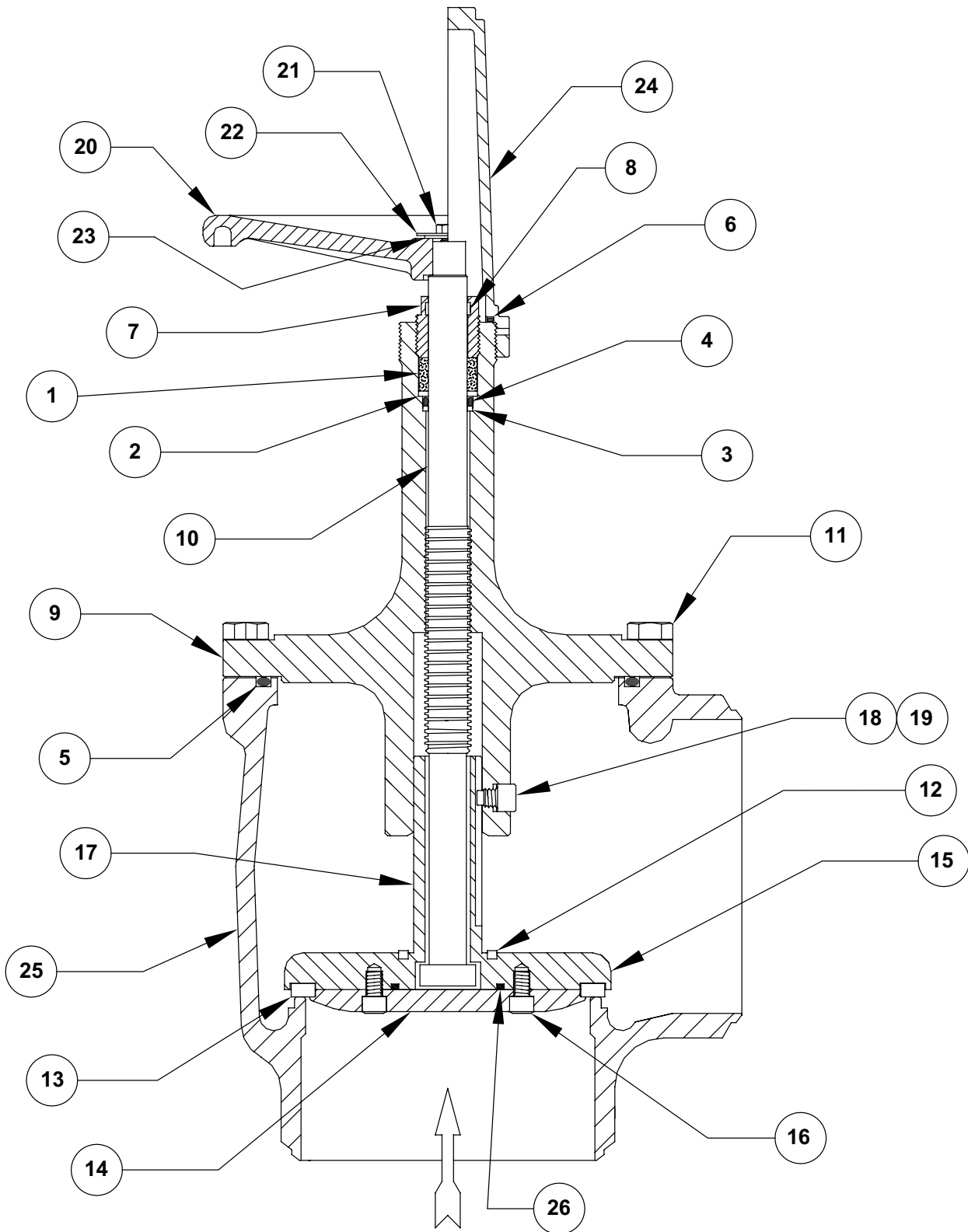
This technical drawing illustrates a mechanical assembly in cross-section. The central component is a vertical shaft. At the top, a flange (14) is secured with a bolt (15) and a nut (17). A seal or gasket (16) is positioned between the flange and the shaft. The shaft passes through a central housing (19) which has a flange (6) and a seal (7). A small component (24) is located near the seal (7). The shaft is supported by a base (8) which has a flange (13) and a seal (20). The base is secured with a bolt (23). The shaft is also supported by a lower housing (9) which has a flange (5) and a seal (10). The lower housing is secured with a bolt (21). A large arrow at the bottom indicates a force or movement applied to the assembly.

G359d
NOV 2019

REPLACEMENT PARTS – BUTT WELD SHUT-OFF VALVES
5" THROUGH 8" (125MM THROUGH 200MM)

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 5" (125mm)		50-1082
	Gasket Kit 6" (150mm)		50-1083
	Gasket Kit 8" (200mm)		50-1084
	<i>Above kits consist of:</i>		
1	Stem Packing	1	
2	Stem Washer	1	
3	Back-up Washer	1	
4	Stem O-Ring	1	
5	Bonnet O-Ring	1	
6	Seal Cap O-Ring	1	
7	Packing Nut	1	
	Bonnet Assembly Kit 5" (125mm)		50-1080
	Bonnet Assembly Kit 6" (150mm)		50-1081
	Bonnet Assembly Kit 8" (200mm)		50-1085
	<i>Above kits consist of:</i>		
8	Bonnet	1	
9	Stem Assembly	1	
13	Bonnet Cap Screw	4	
21	Lower Disc Holder	1	
22	Upper Disc Holder	1	
23	Disc Holder Screw	4	
24	Bushing, Stem Guide	1	
	Gasket Kit	1	
	Disc Assy	1	
	Handwheel Kit 5", 6" (125mm, 150mm)		50-1089
	Handwheel Kit 8" (200mm)		50-1090
	<i>Above kits consist of:</i>		
14	Handwheel	1	
15	Screw	1	
16	Back-Up Washer (8" Only)	1	
17	Nameplate	1	
	Seal Cap Kit 5", 6" (125mm, 150mm)		50-1091
	Seal Cap Kit 8" (200mm)		50-1092
	<i>Above kits consist of:</i>		
6	Seal Cap	1	
19	Seal Cap	1	
20	Body	1	

10" THROUGH 18" (250MM THROUGH 450MM) – BUTT WELD SHUT-OFF VALVES



NOTE: Angle valve shown; kits are the same for globe valves.

REPLACEMENT PARTS – BUTT WELD SHUT-OFF VALVES
10" THROUGH 18" (250MM THROUGH 450MM)

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 10" (250mm)		50-1122
	Gasket Kit 12" (300mm)		50-1123
	Gasket Kit 14" (350mm)		50-1124
	Gasket Kit 16" (400mm)		50-1196
	Gasket Kit 18" (450mm)		50-1197
	<i>Above kits consist of:</i>		
1	Stem Packing	1	
2	Packing Washer	1	
3	Back-up Washer	1	
4	Stem O-Ring	1	
5	Bonnet O-Ring	1	
6	Seal Cap O-Ring	1	
7	Packing Nut	1	
8	Bushing, Stem Guide	1	
	Bonnet Assembly Kit 10" (250mm)		50-1125
	Bonnet Assembly Kit 12" (300mm)		50-1126
	Bonnet Assembly Kit 14" (350mm)		50-1127
	Bonnet Assembly Kit 16" (400mm)		50-1198
	Bonnet Assembly Kit 18" (450mm)		50-1199
	<i>Above kits consist of:</i>		
9	Bonnet	1	
10	Stem Assembly	1	
11	Bonnet Cap Screw	8	
12	Ring, Teflon Back Seat	1	
14	Lower Disc Holder	1	
15	Upper Disc Holder	1	
16	Disc Screw	8	
17	Guide Tube	1	
18	Screw Guide	1	
19	Washer, Screw Guide	1	
	Disc Kit Assy		
	Gasket Kit	1	
	Handwheel Kit 10" (250mm)		50-1131
	Handwheel Kit 12", 14" (300mm, 350mm)		50-1132
	Handwheel Kit 16", 18" (400mm, 450mm)		50-1193
	<i>Above kits consist of:</i>		
20	Handwheel	1	
21	Screw	1	
22	Nameplate	1	
23	Support Washer	1	
	Seal Cap Kit 10" (250mm)		50-1133
	Seal Cap Kit 12", 14" (300mm, 350mm)		50-1134
	Seal Cap Kit 16", 18" (400mm, 450mm)		50-1194
	<i>Above kits consist of:</i>		
6	Seal Cap O-Ring	1	
24	Seal Cap	1	
25	Body	1	

SERVICE AND MAINTENANCE

Hansen steel butt welding shut-off valves require practically no service or maintenance due to the combination of polished stainless steel stems and reliable O-ring stem seals plus graphite composite packing. This almost entirely eliminates stem leakage, the common ailment of shut-off valves.

To help ensure safety, verify the tightness of the packing nut whenever the position (open or closed) is changed on isolation shut-off valves before opening the system. Ensuring that the packing nut is tight helps reduce the possibility that any line or system vibration may cause a slight unseating of a closed valve.

STEM PACKING

When verifying the tightness of the packing nut, use an adjustable wrench. Extrusion of some black graphite packing material along the stem is normal. If the O-ring or the adjustable packing ever needs replacement as evidenced by refrigerant or oil leakage at the stem, open the valve stem firmly to the back-seat position. This separates the O-ring and packing from the system refrigerant. Remove the packing nut carefully and then use a wire hook or a small blade screwdriver to remove the packing and O-ring. Take care not to scratch the stem or bonnet sealing surfaces. Carefully install the backup washer (1½"-18" or 13mm-450mm only), new lubricated stem O-ring, stem washer, and stem packing. Tighten the packing nut only enough to give the handwheel slight turning friction.

VALVE SEAT

To inspect or replace the valve seat disc, isolate the valve from the system and safely pump out all refrigerant to zero pressure with the stem open at least one turn. Evenly loosen all bolts one to two turns. Using a screwdriver, break the seal between the bonnet and valve body, proceeding cautiously to avoid any refrigerant which may still remain inside the valve body. Remove the bonnet bolts and bonnet assembly, being careful not to damage the Teflon seat disc surface. If the seat surface in the body is marred, it may be possible to repair the seat by polishing with emery paper.

If the Teflon seat disc is damaged, replace the entire disc assembly ½" (13mm) thru 4" (100mm) by first removing the ball retainer spring, ball bearings and anti-spin O-ring. Install a new disc assembly including anti-spin O-ring. Alternately, use a lathe to take a ⅙" (0.4 mm) by 45° surface cut on the Teflon seat. The 5" (125mm) thru 18" (450mm) Teflon seats can be replaced by disassembling the disc holder by loosening and removing the disc screws. Replace the Teflon ring, the upper disc O-ring in 10" (250mm) and above, and reassemble the disc.

Replace body gasket or O-ring and reassemble bonnet into body using care not to damage Teflon seat surface. Be careful not to pinch the O-ring. If necessary, retain O-ring in O-ring groove by using a suitable O-ring grease. Ensure the stem is opened at least several turns.

Hansen assembles valves with bonnet cap screws factory tightened as follows: ½" to 1¼" (13mm to 32mm)–30 ft-lb (41 Nm); 1½" (40mm) and 2" (50mm)–40 ft-lb (54 Nm); 2½" (65mm) and 3" (80mm)–60 ft-lb (81 Nm); 4" (100mm)–180 ft-lb (244 Nm); and 5" to 18" (125mm to 450mm)–200 ft-lb (271 Nm). Test the valve for leaks before returning it to service.

TROUBLESHOOTING VALVE LEAKS

On the rare occasion when a small valve leak is discovered, the following procedures may help:

Stem seal leaks – The possibility of this occurrence is based on many factors, such as frequency of use, exercising, debris or rust on stem, etc. First, simply tighten the packing nut to see if this eliminates the leak. If not, back seat the valve or isolate it entirely from the refrigeration system. See Stem Packing section of this bulletin.

Bonnet gasket leaks – If a gasket leak is discovered at the bonnet-to-body interface, first bring the pressure in the valve to zero pressure and check the bonnet bolt tightness. If the leak persists, isolate the valve from the refrigeration system and disassemble the valve. Check the gasket sealing surfaces on the bonnet and valve body for scratches or gouges. These scratches can sometimes be polished smooth. Install a new gasket and reassemble the valve. Evenly tighten all bolts to properly seat the bonnet. Repeat the pressure test.

Bonnet O-ring leaks – Valves 5" (125mm) and larger use bonnet O-rings. If a leak is discovered at the bonnet-to-body interface, first bring the pressure in the valve to zero pressure and check the bonnet bolt torque. The torque recommendations are provided below. If the leak persists, isolate the valve from the refrigeration system and disassemble the valve. Check for debris on the surfaces of the bonnet and O-ring groove in the valve body. Check for scratches and polish the surfaces. Clean and check the O-ring for cuts, cracks and embedded debris. Check the bonnet and valve surface for flatness. Occasionally, the edge of the bonnet on the valve body is struck or dropped which could damage or raise the sealing surface of the bonnet. This prevents the two sealing surfaces from being completely engaged. Remove the raised edge. Also, if not carefully re-assembled, the bonnet may have been slightly askew, preventing proper sealing. Before reassembling the valve, clean the sealing surfaces and O-ring apply suitable grease to the O-ring. Install the bonnet, making sure the stem is open several turns. Evenly tighten, then torque the bonnet bolts to the specified value below. Pressure test the valve to check for leaks.

ORDERING INFORMATION, EXTENDED NECK BUTT WELD SHUT-OFF VALVES

SIZE	DESCRIPTION	CAT NO
1/2" (13mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE051H AWE050H GWE051C AWE050C
3/4" (20mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE076H AWE075H GWE076C AWE075C
1" (25mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE101H AWE100H GWE101C AWE100C
1 1/4" (32mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE126H AWE125H GWE126C AWE125C
1 1/2" (40mm)	Globe, Handwheel Angle Handwheel Globe, Seal Cap Angle, Seal Cap	GWE150H AWE150H GWE150C AWE150C
2" (50mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE200H AWE200H GWE200C AWE200C
2 1/2" (65mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE251H AWE251H GWE251C AWE251C
3" (80mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE301H AWE301H GWE301C AWE301C
4" (100mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWE400H AWE402H GWE400C AWE402C

*Note: 4" Extended Neck Globe is a "T" style valve.

ORDERING INFORMATION, BUTT WELD SHUT-OFF VALVES

SIZE	DESCRIPTION	CAT NO
1/2" (13mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWB051H AWB050H GWB051C AWB050C
3/4" (20mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWB076H AWB075H GWB076C AWB075C
1" (25mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWB101H AWB100H GWB101C AWB100C
1 1/4" (32mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GWB126H AWB125H GWB126C AWB125C
1 1/2" (40mm)	Globe, Handwheel Angle Handwheel Globe, Seal Cap Angle, Seal Cap	GW150H AW150H GW150C AW150C
2" (50mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW200H AW200H GW200C AW200C
2 1/2" (65mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW251H AW251H GW251C AW251C
3" (80mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW301H AW301H GW301C AW301C
4" (100mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW402H AW402H GW402C AW402C
5" (125mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW500H AW500H GW500C AW500C
6" (150mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW600H AW600H GW600C AW600C
8" (200mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW800H AW800H GW800C AW800C
10" (250mm)	Globe, Handwheel Angle Handwheel Globe, Seal Cap Angle, Seal Cap	GW1000H AW1000H GW1000C AW1000C
12" (300mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW1200H AW1200H GW1200C AW1200C
14" (350mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GW1400H AW1400H GW1400C AW1400C
16" (400mm)	Angle, Handwheel Angle, Seal Cap	AW1600H AW1600C
18" (450mm)	Angle, Handwheel Angle, Seal Cap	AW1800H AW1800C

TO ORDER

Specify catalog number. To order valves for 600 psig (41 bar g) SWP, add the letter "C" prefix to catalog number (Example: CGW200H). CE marked valves are available for 1 1/4" through 14" (32mm to 350mm), add the letter "E" suffix (Example: AW150HE).

CAUTION

Hansen valves are for refrigeration only. These instructions must be completely read and understood before selecting, using or servicing Hansen valves and electronics. Only knowledgeable, trained refrigeration mechanics should install, operate, or service. Stated temperature and pressure limits should not be exceeded. Bonnets should not be removed from valves unless system has been evacuated to zero pressure. See also Safety Precautions in the current List Price Schedule and the Safety Precautions Sheet supplied with product.

WARRANTY

All Hansen products, except electronics, are guaranteed against defective materials or workmanship for one year F.O.B. factory. Electronics are guaranteed against defective materials or workmanship for 90 days F.O.B. factory. No consequential damages or field labor is included.

TYPICAL SPECIFICATIONS

"Weldable refrigerant shut-off valves shall have stainless steel stems with dual seals, forged or cast steel bodies, back-seating design for packing replacement, bonnet threads for installation of stem seal caps, butt weld ends machined dimensionally correct for schedule 80 pipe (sizes 1/2"-1 1/2"), schedule 40 pipe (sizes 2"-10") and gauge or standard weight pipe (sizes 12" and above), and suitable for a safe working pressure of 400 psig (28 bar g) or 600 psig (41 bar g) as manufactured by Hansen Technologies Corporation, or approved equal."



Hansen Technologies Corporation
681 Commerce St

Burr Ridge, Illinois 60527 USA

Tel: 630.325.1565 Fax: 630.325.1572 Toll: 866.4HANSEN

Email: sales@hantech.com Web: www.hantech.com

USA • Asia • Europe • India • Latin America • Middle East

© 2013 Hansen Technologies Corporation



GS200H Socket Weld Shut-Off Valve

INTRODUCTION

These low pressure drop refrigerant shut-off valves are designed to be socket-welded directly to steel piping, thereby eliminating potential leaky flanges or threaded joints and simplifying installation. Forged and cast steel bodies are lightweight, yet have substantial wall thickness to help overcome corrosion loss. The cast steel body is rigid, reducing the potential for seat leakage due to flexing of the valve body under abnormal conditions. The “heart” of Hansen shut-off valves is the patented non-leak seal plus packing design. This seal design is used exclusively on Hansen shut-off valves and virtually eliminates stem seal leakage.

APPLICATIONS

Typical uses include:

Ammonia refrigeration system suction, liquid, discharge, recirculating liquid, hot gas, thermosyphon, and oil lines, using handwheel or seal cap models.

Steel pipe portions of halocarbon, commercial, industrial, and air conditioning systems, using seal cap models.

Valves are also available for other fluids such as propane, propylene, and glycol with compatible seals. Contact factory with specific fluid and application details for written approval.

The low friction, non-leak stem seal design permits the use of chain actuation for crossover applications without the need to retighten the packing.

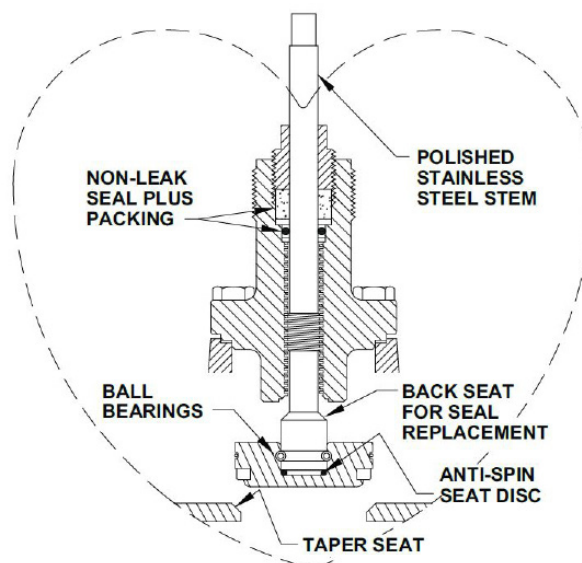
Specifications, Applications, Service Instructions & Parts

SOCKET WELD SHUT-OFF VALVES

1/2" through 2-1/2"
(13 mm through 65 mm)
Globe & Angle
for Refrigerants

CRN CE
Canadian Registration

KEY FEATURES



ADDITIONAL FEATURES

Globe and angle valves available.

Low pressure drop design.

Interchangeable handwheel or seal cap.

Teflon seat disc.

Also available as Hand Expansion Valve (see Bulletin G510).

Chain actuators available 1½"–2½" (40mm–65mm).

Suitable for R717 (ammonia), R22, R134a, R507, R404a, R744 (CO₂) and other Hansen approved refrigerants.

Valves for 600 psig (41 bar g) SWP available.

Extended neck available for additional insulation clearance.

MATERIAL SPECIFICATIONS

Body: ½" to 1¼" (13mm to 32mm), forged steel, ASTM A105; 1½" to 2½" (40mm to 65mm), cast steel, ASTM A-352, Grade LCB

Bonnet: Ductile iron, ½" to 2½" (13mm to 65mm), ASTM A-536

Stem: Polished stainless steel

Disc Holder: Plated steel

Seat Disc: Retained PTFE Teflon

Ball Bearings: Stainless steel

Packing Nut: Zinc plated steel

Stem Packing: Graphite composite plus neoprene O-ring in series

Handwheel: ½" to 1¼" (13mm to 32mm), Zinc plated alloy; 1½" to 2½" (40mm to 65mm), Zinc plated iron alloy

Seal Cap: ½" to 1¼" (13mm to 32mm), Glass-filled polymer, safety vented; 1½" to 2½" (40mm to 65mm), Zinc plated steel

Operating Temperature:

Standard, -60°F to 240°F (-51°C to 115°C), temperatures below -60°F (-51°C) to -76°F (-60°C) at coincident pressures

Optional, -20°F to 300°F (-29°C to 149°C) contact factory

Safe Working Pressure: 400 psig (28 bar g) standard, 600 psig (41 bar g) upon request

Connection Dimensions: ½" to 1½" Match Schedule 80 pipe, 2" to 2½" Match Schedule 40 pipe

ADVANTAGES

Compared to conventional stem seal designs, Hansen's patented non-leak seal plus stem packing virtually eliminates stem seal leakage. In addition, the stem is polished to avoid packing wear. The precision stem threads avoid stem wobble and the packing nut is close fitting to remove dirt particles and frost from the stem. This results in infrequent maintenance or tightening and almost no refrigerant loss.

Hansen valves are designed with anti-spin seat discs. High velocity gas can sometimes cause ordinary seat discs to fail over time. Hansen adds a snubber to prevent the seat disc from freely rotating when valve is in the open position.

Compared to threaded valves, Hansen welding valves eliminate the chance of future leaks at pipe threads. In addition, a socket weld pipe-to-valve body joint eliminates the inherent weakness and corrosion vulnerability of the threaded portion of pipe immediately adjacent to a screwed valve body or flange.

Compared to ball valves, Hansen welding valves have no threaded or flanged-gasket pipe joints. Hansen valves also will not open or close so rapidly as to cause severe pipe shock due to sudden change in liquid velocity. In addition, ball valve stem packing typically cannot be replaced while the valve is in a pressurized line. The stem packing on these Hansen valves can be replaced while the valve is in a pressurized line. Also, dirt or damage to ball valve Teflon seats cannot be overcome by greater stem closing forces, an advantage of Hansen valves.

INSTALLATION

All Hansen weld valves can be installed in horizontal or vertical pipe lines. Stems are normally installed horizontally, but, depending on the application, stems may be installed vertically. Inlet pressure or direction of flow for all valve sizes should normally be under valve seat disc. However, to avoid installing an angle valve with the stem down, it is better to install the valve with the normal flow opposite the direction of the arrow.

Care must be taken when handling and installing large valves. Proper lifting devices and safety precautions must be observed.

A valve may have its bonnet assembly removed before welding. The valve stem should be several turns open when removing and replacing the bonnet assembly. The Teflon seat disc should be protected when outside of the valve. Do not allow the Teflon seat to bump the valve body when removing it or stand the bonnet on the seat disc. Where it is necessary or standard practice to weld a valve into the line without bonnet removal, the stem should be opened several turns to prevent heat damage to the seat disc.

Welds should be annealed as necessary in accordance with good practice. Painting of valves and welds is recommended for corrosion protection. Pipe covering, where applied, should have a proper moisture barrier.

Before putting valves into service, all pipe weld connections, valve seats, bonnet seals, and stem seals should be tested for leaks at pressure levels called for in appropriate codes.

Shut-off valves leading to the atmosphere must not be left unsupervised and must be plugged or capped to prevent corrosion inside the valve as well as leakage due to seat expansion, vibration, pressure shock, or improper opening. Where hand wheel valves are subject to excessive vibration or liquid shock loads, the packing nut should be tightened to prevent handwheel vibration from potentially rotating valve stem. The valve seat should be cracked open to prevent hydrostatic expansion between the valve and the cap. A bleed valve and gauge should be installed to provide a means of monitoring and pump-out. Valves should never directly feed a water tank because of potential internal corrosion or seat opening caused by vibration.

INSULATION

Conventional valve-shaped block insulation can occasionally be used for both angle and globe valves. However, fabricated insulation shapes are recommended. If not available locally, Hansen can recommend a source of high quality, economical valve insulation. For exterior valve insulation dimensions, refer to Sales Drawing 5002-28.

EXTENDED NECK VALVES

Hansen bonnets with extended necks, ½" thru 2½" (13mm–65mm), meet or exceed the recommended low temperature insulation thickness dimensions of the IIAR Piping Handbook. In addition, the stem and packing nut trim are upgraded to 316 for stainless steel materials. Stainless steel trim is recommended for environments where severe corrosive conditions exist. (Add an "E" suffix to the model number, for example ASE200C or GSE150H.)

SIZING GUIDE

These flow capacity recommendations are not affected by the length of the pipe line. These are approximate optimum sizes based on power costs versus the investment costs of piping and its total installed cost. Piping sized to this capacity will have a 1°F (0.6°C) pressure drop for the following equivalent lengths:

suction lines 700 diameters

discharge lines 1400 diameters

liquid lines 2400 diameters

Example: Hansen valves have about 145 diameters of equivalent flow resistance, or $145/700 = 0.2^\circ\text{F}$ (0.12°C) of equivalent pressure drop at the suction line capacities shown for a valve in a suction line.

The rationale for the vapor line sizing was developed by William V. Richards in two papers: "Refrigerant Vapor Line Sizing Not Dependent of Length," 16th International Congress of Refrigeration, IIR, Paris, 1983; and "Practical Pipe Sizing for Refrigerant Vapor Lines," Sixth Annual Meetings, IIAR, San Francisco, 1984.

FLOW COEFFICIENTS

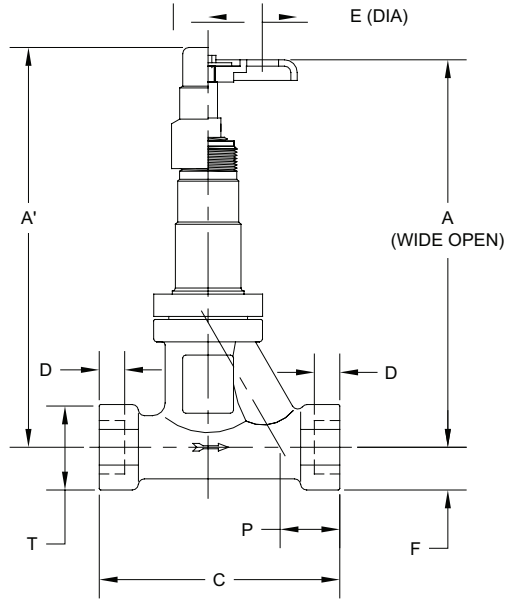
NOMINAL SIZE		ANGLE				GLOBE			
INCH	MM	FLOW COEFFICIENT		EQUIVALENT LENGTH		FLOW COEFFICIENT		EQUIVALENT LENGTH	
		Cv	Kv	FEET	METERS	Cv	Kv	FEET	METERS
1/2"	13	6.0	5.2	3.7	1.1	4.0	3.5	8.4	2.6
3/4"	20	9.0	7.8	8.4	2.6	8.0	6.9	11	3.2
1"	25	19	16	7.2	2.2	15	13	12	3.5
1¼"	32	21	18	27	8.2	16	14	46	14
1½"	40	53	46	10	3.0	41	35	16	5.0
2"	50	80	69	23	7.0	67	58	33	10
2½"	65	173	150	13	3.8	163	141	14	4.3

FLOW CAPACITIES PIPING AND VALVE SIZING GUIDE FOR AMMONIA

SERVICE	CONDITIONS		CAPACITIES (MM)							
	TEMPERATURE °F (°C)	PRESSURE PSIG (BAR)	1/2" (13) TONS (kW)	3/4" (20) TONS (kW)	1" (25) TONS (kW)	1¼" (32) TONS (kW)	1½" (40) TONS (kW)	2" (50) TONS (kW)	2½" (65) TONS (kW)	
Suction Lines Single Stage Compressor	+20 (-6.7)	33.5 (2.3)	– –	– –	8.6 (30)	15.8 (56)	21.3 (75.0)	35.7 (126)	51.1 (180)	
	0 (-17.8)	15.7 (1.1)	– –	– –	5.7 (20)	10.4 (37)	13.9 (49.0)	22.7 (80.0)	34.0 (120)	
Suction Lines Booster	-20 (-28.9)	3.6 (0.25)	– –	– –	4.2 (15)	7.4 (26)	10.3 (36.0)	16.8 (59.0)	24.8 (87.0)	
	-40 (-40)	8.7" (0.7)	– –	– –	– –	4.4 (15)	6.3 (22.0)	9.9 (35.0)	14.4 (51.0)	
Liquid Overfeed Return Lines (4X)	+20 (-6.7)	33.5 (2.3)	– –	– –	5.0 (18)	9.1 (32)	12.3 (43.0)	20.6 (72.4)	29.4 (103)	
	0 (-17.8)	15.7 (1.1)	– –	– –	3.4 (12)	6.3 (22)	8.5 (30.0)	13.6 (48.0)	20.5 (72.0)	
	-20 (-28.9)	3.6 (0.25)	– –	– –	2.2 (8.0)	4.0 (14)	5.5 (19.0)	8.9 (31.3)	13.1 (46.0)	
	-40 (-40)	8.7" (0.7)	– –	– –	– –	2.4 (8.0)	3.4 (12.0)	5.4 (19.0)	7.90 (28.0)	
Hot Gas Feed Hot Gas Main	+70 (+21.1)	114 (7.9)	2.2 (8)	4.3 (15)	7.3 (26)	14.1 (50)	19.6 (69.0)	36.5 (128)	53.0 (186)	
	+70 (+21.1)	114 (7.9)	4.4 (15)	8.6 (30)	14.7 (52)	28.1 (99)	39.2 (138)	73.0 (257)	106 (373)	
Compressor Discharge	+86 (+30)	155 (10.7)	– –	– –	12.6 (44)	24.1 (85)	33.6 (118)	62.6 (220)	90.3 (318)	
Condenser Drains	+86 (+30)	– –	6.0 (21)	14.5 (51)	24.0 (84)	50.0 (176)	77.0 (271)	140 (492)	220 (774)	
Liquid Mains	+86 (+30)	– –	28.3 (100)	53.1 (187)	90.8 (320)	143 (503)	202 (711)	454 (1597)	657 (2311)	
Liquid Feed Branch	+86 (+30)	– –	54.9 (193)	103 (363)	176 (620)	277 (975)	392 (1380)	881 (3098)	1273 (4477)	
Liquid Overfeed Supply (4X)	+10 (-12.2)	– –	9.0 (32)	17.0 (60)	29.0 (102)	46.0 (162)	65.0 (229)	144 (506)	208 (732)	

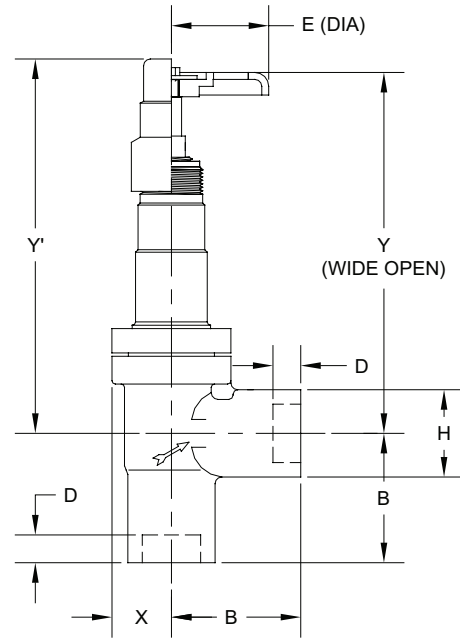
INSTALLATION DIMENSIONS

1/2" THROUGH 2 1/2" (13MM THROUGH 65MM)



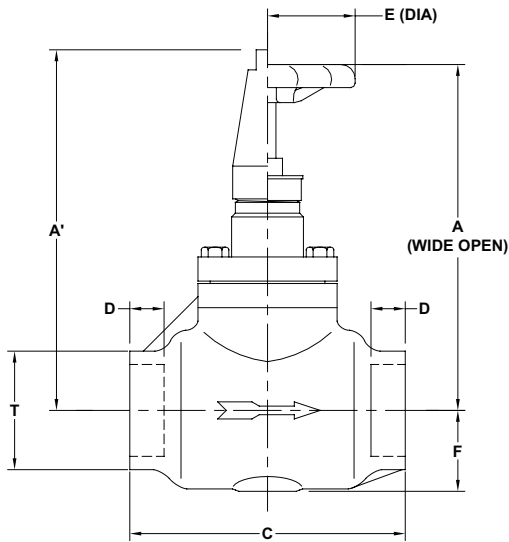
GLOBE

1/2" THROUGH 1 1/4" (13MM – 32MM)



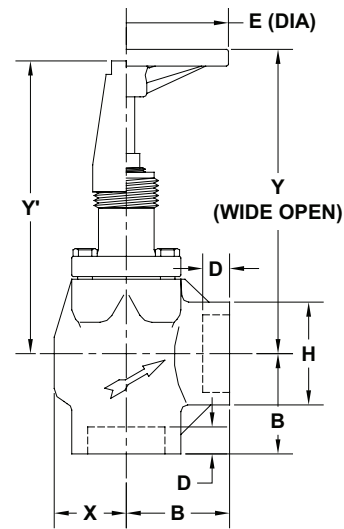
ANGLE

1/2" THROUGH 1 1/4" (13MM – 32MM)



GLOBE

1 1/2" THROUGH 2 1/2" (40MM – 65MM)



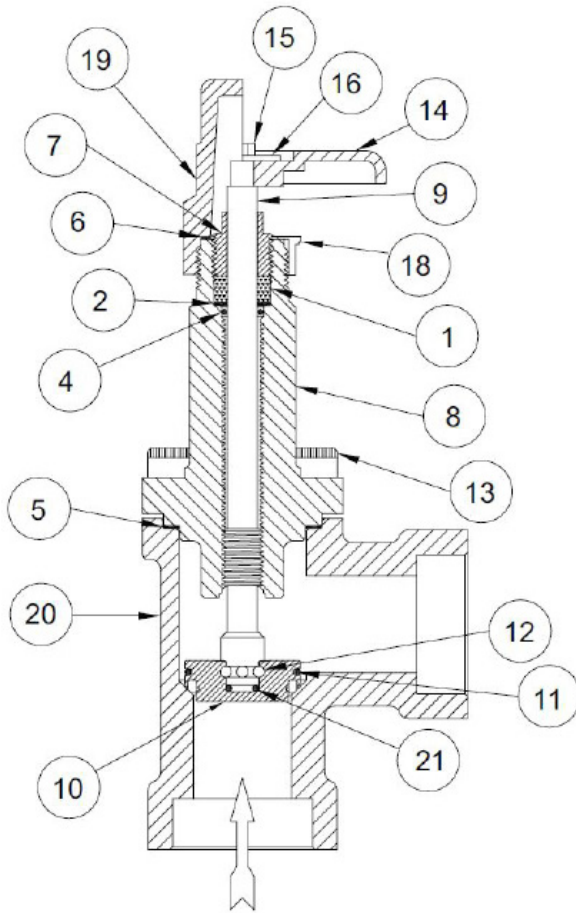
ANGLE

1 1/2" THROUGH 2 1/2" (40MM – 65MM)

SIZE (mm)	GLOBE					ANGLE					D	E	ADD TO A, A', Y, Y' EXT NECK
	A	A'	C	T	F	Y	Y'	B	H	X			
1/2" (13)	7.62" (194)	7.86" (200)	4.73" (120)	1.68" (43)	0.84" (21)	6.46" (164)	6.70" (170)	2.33" (59)	1.57" (40)	1.07" (27)	0.50" (13)	3.50" (89)	1.8" (45)
3/4" (20)	7.62" (194)	7.86" (200)	4.73" (120)	1.68" (43)	0.84" (21)	6.46" (164)	6.70" (170)	2.33" (59)	1.57" (40)	1.07" (27)	0.50" (13)	3.50" (89)	1.8" (45)
1" (25)	7.88" (200)	8.09" (206)	5.93" (150)	2.31" (58)	1.15" (29)	6.24" (158)	6.52" (166)	2.75" (75)	2.31" (59)	1.22" (31)	0.50" (13)	3.50" (89)	1.8" (45)
1 1/4" (32)	7.88" (200)	8.09" (206)	5.93" (150)	2.31" (58)	1.15" (29)	6.24" (158)	6.52" (166)	2.75" (75)	2.31" (59)	1.22" (31)	0.62" (16)	3.50" (89)	1.8" (45)
1 1/2" (40)	8.88" (225)	9.13" (232)	7.25" (184)	3.00" (76)	2.12" (54)	7.75" (197)	8.00" (203)	3.50" (89)	3.00" (76)	1.88" (48)	0.62" (16)	4.25" (108)	2.85" (72)
2" (50)	8.88" (225)	9.13" (232)	7.25" (184)	3.00" (76)	2.12" (54)	7.75" (197)	8.00" (203)	3.50" (89)	3.00" (76)	1.88" (48)	0.88" (22)	4.25" (108)	2.85" (72)
2 1/2" (65)	12.13" (308)	11.75" (298)	9.25" (235)	3.75" (95)	2.75" (70)	11.00" (280)	10.63" (270)	3.50" (89)	3.00" (76)	2.38" (60)	1.00" (25)	7.63" (194)	2.88" (73)

REPLACEMENT PARTS – SOCKET WELD SHUT-OFF VALVES

1/2" THROUGH 1 1/4" (13MM THROUGH 32MM)



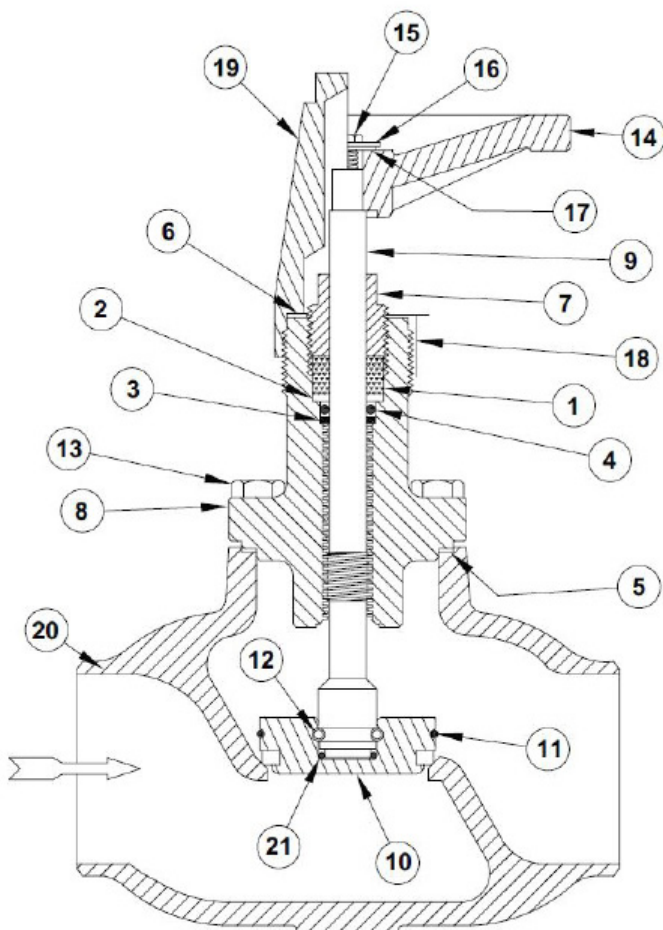
NOTE: Angle valve shown; kits are the same for globe valves.

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 1/2", 3/4", 1", 1-1/4" (13mm, 20mm, 25mm, 32mm)		50-1040
	Gasket Kit, Extended Neck 1/2", 3/4", 1", 1-1/4" (13mm, 20mm, 25mm, 32mm)		50-1171
	<i>Above kits consist of:</i>		
1	Stem Packing	1	
2	Stem Washer	1	
4	Stem O-Ring	1	
5	Bonnet Gasket	1	
6	Seal Cap O-Ring	1	
7a	Packing Nut	1	
7b	Packing Nut, Extended Neck	1	
21	O-Ring, Anti-Spin	1	
*	Bonnet Assembly Kit 1/2", 3/4" (13mm, 20mm)		50-1094
	Bonnet Assembly Ext Neck Kit 1/2", 3/4" (13mm, 20mm)		50-1172
*	Bonnet Assembly Kit 1", 1-1/4" (25mm, 32mm)		50-1095
	Bonnet Assembly Ext Neck Kit 1", 1-1/4" (25mm, 32mm)		50-1174
	<i>Above kits consist of:</i>		
8	Bonnet 1/2", 3/4" (13mm, 20mm)	1	
9a	Stem	1	
9b	Stem, Extended Neck	1	
10	Disc Assembly	1	
11	Ball Retainer	1	
12	Balls	10	
13	Bonnet Cap Screws	4	
	Gasket Kit	1	
	Gasket Kit, Ext Neck	1	
	Handwheel Kit		50-1005
	<i>Above kits consist of:</i>		
14	Handwheel	1	
15	Screw	1	
16	Nameplate	1	
18	Bonnet Thread Cap	1	
	Seal Cap Kit		50-1036
	<i>Above kits consist of:</i>		
6	Seal Cap O-Ring	1	
19	Seal Cap	1	
20	Body		

Note: A plate steel cap is available. To order specify part number 50-1064.

REPLACEMENT PARTS – SOCKET WELD SHUT-OFF VALVES

1½" THROUGH 2½" (40MM THROUGH 65MM)



NOTE: Globe valve shown; kits are the same for angle valves.

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 1-1/2", 2" consists of:		50-1023
1	Stem Packing	1	
2	Stem Washer	1	
3	Back-Up Washer	1	
4	Stem O-Ring	1	
5	Bonnet Gasket	1	
6	Seal Cap Gasket	1	
7	Packing Nut	1	
21	O-Ring, Anti-Spin	1	
	Bonnet Assembly Kit 1-1/2", 2"		50-1024
	<i>Above kits consist of:</i>		
8	Bonnet	1	
9	Stem	1	
10	Disc Assembly	1	
11	Ball Retainer	1	
12	Balls	16	
13	Bonnet Cap Screw	4	
	Gasket Kit	1	
	Handwheel Kit 1-1/2", 2" consists of:		50-1026
14	Handwheel	1	
16	Name Plate	1	
15	Screw	1	
17	Bonnet Thread Cap	1	
	Seal Cap Kit 1-1/2", 2" consists of:		50-1027
19	Seal Cap	1	
6	Seal Cap Gasket	1	
20	Body	1	

ITEM	DESCRIPTION	QTY	PART NO
	Gasket Kit 2-1/2" consists of:		50-1043
1	Stem Packing	1	
2	Stem Washer	1	
3	Back-Up Washer	1	
4	Stem O-Ring	1	
5	Bonnet Gasket	1	
6	Seal Cap Gasket	1	
7	Packing Nut	1	
21	O-Ring, Anti-Spin	1	
	Bonnet Assembly Kit 2-1/2"		50-1044
	<i>Above kits consist of:</i>		
8	Bonnet	1	
9	Stem	1	
10	Disc Assembly	1	
11	Ball Retainer	1	
12	Balls	16	
13	Bonnet Cap Screw	4	
	Gasket Kit	1	
	Handwheel Kit 2-1/2" consists of:		50-1037
14	Handwheel	1	
16	Name Plate	1	
15	Screw	1	
17	Support Washer	1	
	Seal Cap Kit 2-1/2" consists of:		50-1038
19	Seal Cap	1	
6	Seal Cap Gasket	1	
20	Body	1	

SERVICE AND MAINTENANCE

Hansen steel socket welding shut-off valves require practically no service or maintenance due to the combination of polished stainless steel stems and reliable O-ring stem seals plus graphite composite packing. This almost entirely eliminates stem leakage, the common ailment of shut-off valves.

To help ensure safety, verify the tightness of the packing nut whenever the position (open or closed) is changed on isolation shut-off valves before opening the system. Ensuring that the packing nut is tight helps reduce the possibility that any line or system vibration may cause a slight unseating of a closed valve.

STEM PACKING

When verifying the tightness of the packing nut, use an adjustable wrench. Extrusion of some black graphite packing material along the stem is normal. If the O-ring or the adjustable packing ever needs replacement as evidenced by refrigerant or oil leakage at the stem, open the valve stem firmly to the back-seat position. This separates the O-ring and packing from the system refrigerant. Remove the packing nut carefully and then use a wire hook or a small blade screwdriver to remove the packing and O-ring. Take care not to scratch the stem or bonnet sealing surfaces. Carefully install a backup washer, new lubricated stem O-ring, stem washer, and stem packing. Tighten the packing nut only enough to give the handwheel slight turning friction.

VALVE SEAT

To inspect or replace the valve seat disc, isolate the valve from the system and safely pump out all refrigerant to zero pressure with the stem open at least one turn. Evenly loosen all bolts one to two turns. Using a screwdriver, break the seal between the bonnet and valve body, proceeding cautiously to avoid any refrigerant which may still remain inside the valve body. Remove the bonnet bolts and bonnet assembly, being careful not to damage the Teflon seat disc surface. If the seat surface in the body is marred, it may be possible to repair the seat by polishing with emery paper.

If the Teflon seat disc is damaged, replace the entire disc assembly $\frac{1}{2}$ " (13mm) thru $2\frac{1}{2}$ " (65mm) by first removing the ball retainer spring, ball bearings and anti-spin O-ring. Install a new disc assembly including anti-spin O-ring. Alternately, use a lathe to take a $\frac{1}{64}$ " (0.4 mm) by 45° surface cut on the Teflon seat.

Replace body gasket or O-ring and reassemble bonnet into body using care not to damage Teflon seat surface. Be careful not to pinch the O-ring. If necessary, retain O-ring in O-ring groove by using a suitable O-ring grease. Ensure the stem is opened at least several turns.

Hansen assembles valves with bonnet cap screws factory tightened as follows: $\frac{1}{2}$ " to $1\frac{1}{4}$ " (13mm to 32mm)–30 ft-lb (41 Nm); $1\frac{1}{2}$ " (40mm) and 2" (50mm)–40 ft-lb (54 Nm); and $2\frac{1}{2}$ " (65mm)–60 ft-lb (81 Nm). Test the valve for leaks before returning it to service.

TROUBLESHOOTING VALVE LEAKS

On the rare occasion when a small valve leak is discovered, the following procedures may help:

Stem seal leaks – The possibility of this occurrence is based on many factors, such as frequency of use, exercising, debris or rust on stem, etc. First, simply tighten the packing nut to see if this eliminates the leak. If not, back seat the valve or isolate it entirely from the refrigeration system. See Stem Packing section of this bulletin.

Bonnet gasket leaks – If a gasket leak is discovered at the bonnet-to-body interface, first bring the pressure in the valve to zero pressure and check the bonnet bolt tightness. If the leak persists, isolate the valve from the refrigeration system and disassemble the valve. Check the gasket sealing surfaces on the bonnet and valve body for scratches or gouges. These scratches can sometimes be polished smooth. Install a new gasket and reassemble the valve. Evenly tighten all bolts to properly seat the bonnet. Repeat the pressure test.

ORDERING INFORMATION, SOCKET WELD SHUT-OFF VALVES

SIZE	DESCRIPTION	CAT NO
1/2" (13 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GSB051H ASB050H GSB051C ASB050C
3/4" (20 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GSB076H ASB075H GSB076C ASB075C
1" (25 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GSB101H ASB100H GSB101C ASB100C
1 1/4" (32 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GSB126H ASB125H GSB126C ASB125C
1 1/2" (40 mm)	Globe, Handwheel Angle Handwheel Globe, Seal Cap Angle, Seal Cap	GS150H AS150H GS150C AS150C
2" (50 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GS200H AS200H GS200C AS200C
2 1/2" (65 mm)	Globe, Handwheel Angle, Handwheel Globe, Seal Cap Angle, Seal Cap	GS251H AS251H GS251C AS251C

TO ORDER

Specify catalog number. To order valves with extended neck, add the letter "E" to the catalog number (Example: GSE051H or ASE125C). To order valves for 600 psig (41 bar g) SWP, add the letter "C" prefix to catalog number (Example: CGSB076H or CASB100H). CE marked valves are available, add the letter "E" suffix to catalog number for sizes 1 1/4" (32mm) and larger (Example: AS150HE).

CAUTION

Hansen valves are for refrigeration systems only. These instructions must be completely read and understood before selecting, using, or servicing Hansen valves and electronics. Only knowledgeable, trained refrigeration mechanics should install, operate, or service. Stated temperature and pressure limits should not be exceeded. Bonnets should not be removed from valves unless the system has been evacuated to zero pressure. See also Safety Precautions in the current List Price Schedule and the Safety Precautions Sheet supplied with the product.

WARRANTY

All Hansen products, except electronics, are guaranteed against defective materials or workmanship for one year F.O.B. factory. Electronics are guaranteed against defective materials or workmanship for 90 days F.O.B. factory. No consequential damages or field labor is included.

TYPICAL SPECIFICATIONS

"Weldable refrigerant shut-off valves shall have stainless steel stems with dual seals, forged or cast steel bodies, back-seating design for packing replacement, bonnet threads for installation of stem seal caps, socket weld ends machined dimensionally correct for schedule 80 pipe (sizes 1/2" to 1 1/2") and schedule 40 pipe (sizes 2" to 2 1/2"), and suitable for a safe working pressure of 400 psig (28 bar g) or 600 psig (41 bar g), as manufactured by Hansen Technologies Corporation, or approved equal."



Hansen Technologies Corporation
400 Quadrangle Drive, Suite F
681 Commerce St
Burr Ridge, Illinois 60527 USA

: 866.4HANSEN

Email: sales@hantech.com Web: www.hantech.com

USA • Asia • Europe • India • Latin America • Middle East

© 2013 Hansen Technologies Corporation